

Dr. Rania Abdelghani

Education

Inria University of Bordeaux <i>PhD in Cognitive science</i>	<i>Sept. 2024</i>
◦ Advisors: Prof. Dr. Pierre-Yves Oudeyer and Prof. Dr. Hélène Sauzéron	
◦ Title: Conversational agents to train curiosity and metacognition in young learners.	
Sorbonne University, Paris <i>MSc. in Engineering sciences</i>	<i>Sept. 2020</i>
INSAT University, Tunis <i>M.Eng in intelligent systems</i>	<i>Sept. 2019</i>

Scientific Experience

Postdoctoral researcher <i>Hector Research Institute University of Tuebingen</i>	<i>Tübingen, DE</i> <i>November 2024- Now</i>
◦ Advisor: Prof. Dr. Kou Murayama	
Host Lecturer <i>University of Côte d'Azur</i>	<i>Nice, FR</i> <i>Sept. 2024- Dec. 2024</i>
Visiting Researcher <i>University of Berkeley - Kidd Lab</i>	<i>California, USA</i> <i>Sept. 2023- Dec. 2023</i>
◦ Host: Prof. Dr. Celeste Kidd	
Contractual lecturer <i>University of Bordeaux</i>	<i>Bordeaux, FR</i> <i>Sept. 2022 – Jul. 2023</i>
Research engineer <i>EvidenceB</i>	<i>Paris, FR</i> <i>Sept. 2020 – Dec. 2020</i>
Research intern <i>ISIR Institute</i>	<i>Paris, FR</i> <i>Jan. 2020 – Sept. 2020</i>
Research engineer intern <i>Therapanacea</i>	<i>Paris, FR</i> <i>Jan. 2019 – Sept. 2019</i>

Mentoring

As a graduate student at the University of Bordeaux, I co-supervised :

Carole Morvan and Lucas Spooner <i>Masters 1 students in cognitive sciences - 3-month internships</i>	<i>May. 2024- Jul. 2024</i>
Chloé Desvaux (PhD student in psychology, University of Bordeaux) <i>Masters 2 student in cognitive sciences - 6-month internship</i>	<i>Jan. 2023- Jun. 2023</i>
Pauline Lucas <i>Masters 2 student in cognitive sciences - 6-month internship</i>	<i>Jan. 2022- Jun. 2022</i>
Cléa Gardin <i>Masters 1 student in cognitive sciences - 3-month internship</i>	<i>Jun. 2021- Sept. 2021</i>

Grants & Awards experiences

LEAD Intramural Grant - Accepted <i>Investigating students' strategies when using LLMs during mathematical learning</i> € 5000	<i>July 2025</i>
Eliteprogramm für Postdocs Grant - Rejected	<i>April 2025</i>

Bordeaux mobility scholarship (UBGRS Mob) Grant - Accepted
€ 4,500

May 2023

Best presentation award at the "European Advanced Educational technologies" conference - Accepted

March 2022

Publications

A full list is available on [Google Scholar](#) 

- **Abdelghani, R.**, Sauzéon, H., Murayama, K., Kidd, C., & Oudeyer, P.Y. (2025). Students fail to ask in-context questions with ChatGPT and to accurately evaluate its answers. (In revision).
- Laak, K. J., **Abdelghani, R.**, & Aru, J. (2024, July). Personalisation is not Guaranteed: The Challenges of Using Generative AI for Personalised Learning. In International conference on innovative technologies and learning (pp. 40-49). Cham: Springer Nature Switzerland.
- **Abdelghani, R.** (2024). Guiding the minds of tomorrow: conversational agents to train curiosity and metacognition in young learners (Doctoral dissertation, Université de Bordeaux).
- **Abdelghani, R.**, Sauzéon, H., & Oudeyer, P. Y. (2023). Generative AI in the Classroom: Can Students Remain Active Learners?. arXiv preprint arXiv:2310.03192.
- **Abdelghani, R.**, Law, E., Desvaux, C., Oudeyer, P. Y., & Sauzéon, H. (2023, June). Interactive environments for training children's curiosity through the practice of metacognitive skills: a pilot study. In Proceedings of the 22nd Annual ACM Interaction Design and Children Conference (pp. 495-501).
- Yuan, X., Wang, T., Wang, Y. H., Fine, E., **Abdelghani, R.**, Lucas, P., ... & Oudeyer, P. Y. (2022). Selecting better samples from pre-trained llms: A case study on question generation. arXiv preprint arXiv:2209.11000.
- **Abdelghani, R.**, Wang, Y. H., Yuan, X., Wang, T., Lucas, P., Sauzéon, H., & Oudeyer, P. Y. (2024). GPT-3-driven pedagogical agents to train children's curious question-asking skills. International Journal of Artificial Intelligence in Education, 34(2), 483-518.
- Xiao, Z., Yuan, X., Liao, Q. V., **Abdelghani, R.**, & Oudeyer, P. Y. (2023, March). Supporting qualitative analysis with large language models: Combining codebook with GPT-3 for deductive coding. In Companion proceedings of the 28th international conference on intelligent user interfaces (pp. 75-78).
- **Abdelghani, R.**, Oudeyer, P. Y., Law, E., de Vulpillières, C., & Sauzéon, H. (2022). Conversational agents for fostering curiosity-driven learning in children. International Journal of Human-Computer Studies, 167, 102887.
- Stower, R., **Abdelghani, R.**, Tschopp, M., Evangelista, K., Chetouani, M., & Kappas, A. (2022). Exploring space for robot mistakes in child robot interactions. Interaction Studies, 23(2), 243-288.

Conference talks and workshops

Some oral presentations:

Webinair "Cogni'classes"

Question-asking in the digital classroom: theoretical aspects and applications

Paris, France

Jan. 2025

International Conference on Artificial Intelligence in Education

Generative AI for promoting active learning strategies

Recife, Brasil

Jul. 2024

GenAI in Education Workshop - Neural Information Processing Systems

Generative AI in the classroom: can students remain active learners

New Orleans, USA

Dec. 2023

Interaction Design and Children Conference

interactive environments for training children's curiosity through the practice of metacognitive skills : a pilot study.

New Orleans, USA

Jun. 2023

Budapest CEU Conference on Cognitive Development

An interactive environment for training children's curiosity-driven learning through

Budapest, Hungary

Jan. 2023

metacognition.

Interaction Design and Children Conference

Role of curiosity-driven design in promoting sustainable learning technologies.

Braga, Portugal

Jan. 2022

European Advanced Educational Technology Conference

Conversational agents for training children's curiosity-driven behavior

Cambridge, UK

March. 2022

Participation in workshop organization:

Independent workshop

Workshop: Designing assistive technology

Bordeaux, France

Sept. 2021

Independent workshop

Workshop: Inquiry and creativity in childhood: what role for new technologies?

Bordeaux, France

Sept. 2024

Budapest CEU Conference on Cognitive Development

Symposium: Truth, lies, and misinformation during cognitive development

Budapest, Hungary

Jan. 2023

Skills

- **Disciplines:** Cognitive science, experimental cognitive psychology, educational psychology, AI for education, Human-centered AI, Human-centered design, Natural Language Processing and prompt engineering
- **Coding:** Python, R, C, HTML, CSS
- **Tools:** Bash, Git
- **Operating systems:** Linux
- **Typesetting:** $\text{\LaTeX}$
- **Languages:** Native in Arabic, Proficient in English and French (C2), Beginner in Spanish and German (A1).